

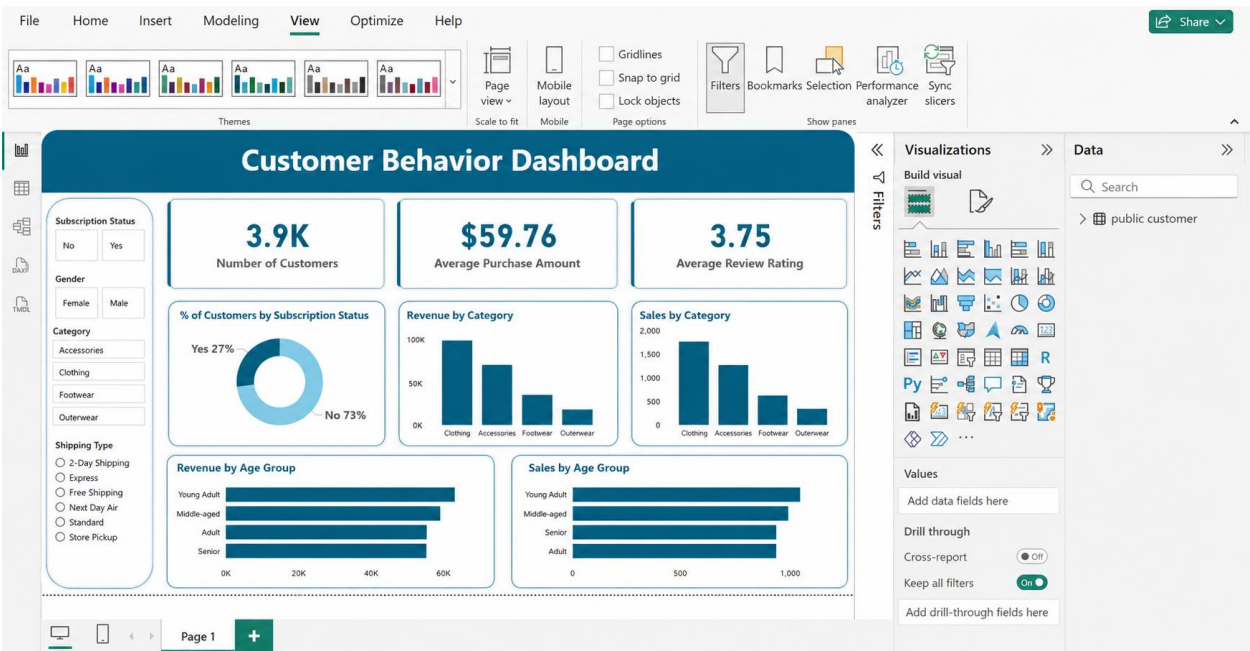
Consumer Behavior Analytics Dashboard Report

Executive Summary

Understanding consumer purchasing behaviour is essential for retailers seeking to improve customer satisfaction, maximise revenue, and optimise marketing strategies. This report presents an analysis of a retail consumer dataset containing **3,900 customer transactions**. The dashboard was developed using **Power BI** after performing data cleaning and preprocessing in Python.

The dashboard highlights key business indicators including customer count, average purchase amount, customer review ratings, subscription status, revenue by product category, sales by category, and purchasing behaviour across different age groups. Interactive filters enable business users to analyse purchasing behaviour based on customer demographics and shopping preferences.

Overall, the analysis indicates that **Clothing** is the highest-performing product category, subscription customers represent approximately **27%** of the customer base, and the average customer spends **\$59.76** per transaction while providing an average review rating of **3.75 out of 5**.



1. Introduction

Retail businesses generate large volumes of transactional data every day. Analysing this information enables organisations to understand customer preferences, identify profitable product categories, and make informed business decisions.

The objective of this project was to build an interactive Business Intelligence dashboard capable of summarising customer purchasing behaviour and providing management with meaningful insights into sales performance and customer segmentation.

The dashboard was developed using Power BI after data preprocessing in Python, including cleaning missing values, standardising column names, and preparing calculated measures for analysis.

2. Dataset Overview

The dataset contains information for **3,900 retail customers** with attributes describing demographics, purchasing habits, shopping preferences, and transaction details.

Dataset Characteristics

Attribute	Description
Total Records	3,900 Customers
Total Variables	18
Customer Demographics	Age, Gender, Location
Product Information	Category, Item Purchased, Colour, Size
Shopping Behaviour	Previous Purchases, Frequency, Shipping Type
Customer Engagement	Subscription Status, Discount Usage, Promo Code
Financial Metrics	Purchase Amount
Customer Satisfaction	Review Rating

3. Dashboard Overview

The dashboard was designed to provide management with an easy-to-understand overview of customer purchasing behaviour.

The dashboard consists of:

- Three KPI cards
- Two category performance charts
- Two demographic analysis charts
- One subscription distribution chart
- Interactive slicers for customer segmentation

The dashboard allows filtering by:

- Subscription Status
- Gender
- Product Category
- Shipping Type

This enables decision-makers to analyse customer behaviour from multiple business perspectives.

4. Key Performance Indicators (KPIs)

4.1 Number of Customers

Value: 3,900

The dataset contains purchase information from **3,900 customers**, providing a sufficiently large sample for analysing shopping behaviour and identifying purchasing trends.

4.2 Average Purchase Amount

Value: \$59.76

Customers spend an average of **\$59.76** per transaction, indicating a moderate transaction value. This metric can be used as a benchmark when evaluating promotional campaigns or pricing strategies.

4.3 Average Review Rating

Value: 3.75 / 5

The average customer review rating of **3.75** suggests that customers are generally satisfied with their purchases. However, there remains room for improving customer experience to increase overall satisfaction.

5. Customer Subscription Analysis

The subscription status chart shows the distribution of customers between subscribed and non-subscribed shoppers.

Findings

- **73%** of customers are **non-subscribers**
- **27%** are subscribed customers

Business Insight

The relatively low subscription rate indicates a significant opportunity for the business to expand loyalty or membership programs. Increasing subscription adoption could improve customer retention, encourage repeat purchases, and strengthen long-term customer relationships.

6. Revenue Analysis

Revenue has been analysed across different product categories.

Key Findings

The revenue chart indicates that:

1. **Clothing** generates the highest revenue.
2. **Accessories** are the second-highest revenue contributor.
3. **Footwear** contributes a moderate share.
4. **Outerwear** produces the lowest revenue among the categories.

Business Insight

The Clothing category appears to be the company's primary revenue driver. Maintaining inventory availability and expanding the clothing product range may further increase overall sales. Marketing efforts could also focus on improving sales within the lower-performing Outerwear category.

7. Sales Performance by Category

The sales distribution closely follows the revenue trends.

The dashboard indicates:

- Clothing records the highest sales volume.
- Accessories maintain strong customer demand.
- Footwear attracts fewer purchases.
- Outerwear receives the least customer demand.

Business Insight

Because both sales volume and revenue favour Clothing, this category demonstrates consistent customer demand rather than relying solely on higher-priced products.

8. Revenue by Age Group

Customer spending varies across different age groups.

The dashboard shows that:

- **Young Adults** generate the highest revenue.
- **Middle-aged** customers closely follow.
- **Adults** contribute moderate revenue.
- **Senior** customers generate the lowest revenue.

Business Insight

Young adults appear to be the most valuable customer segment. Marketing campaigns, loyalty rewards, and personalised recommendations should primarily target this demographic to maximise future revenue.

9. Sales by Age Group

Purchase frequency follows a similar pattern.

Young adults complete the greatest number of purchases, followed by middle-aged customers. Adults and seniors record comparatively lower transaction volumes.

Business Insight

This suggests that younger consumers are both higher-spending and more frequent shoppers, making them an ideal target market for promotional campaigns and new product launches.

10. Interactive Dashboard Features

The dashboard includes multiple interactive filters that improve business analysis.

Users can filter dashboard results using:

- Subscription Status
- Gender
- Product Category
- Shipping Type

These filters enable management to compare customer segments and identify purchasing patterns across different demographic groups without modifying the underlying dataset.

11. Business Recommendations

Based on the dashboard findings, the following recommendations are proposed:

Increase Subscription Adoption

Since only **27%** of customers are subscribed, introducing stronger membership benefits and exclusive discounts could improve customer retention and encourage repeat purchases.

Focus on Clothing Products

Clothing generates both the highest sales volume and revenue. Increasing inventory, expanding product offerings, and prioritising promotional activities in this category may further improve profitability.

Target Young Adults

Young adults contribute the largest share of revenue and purchases. Digital marketing campaigns and personalised promotions should primarily target this customer segment.

Improve Low-Performing Categories

Outerwear consistently records the lowest sales and revenue. Product assortment, pricing strategies, and promotional offers should be reviewed to improve performance.

Enhance Customer Satisfaction

Although the average review rating of **3.75** indicates generally positive customer experiences, improving product quality, delivery speed, and customer service could increase customer satisfaction and strengthen brand loyalty.

12. Conclusion

The Consumer Behavior Dashboard provides a comprehensive overview of retail purchasing behaviour using interactive visualisations and key business metrics.

Analysis of **3,900 customer transactions** demonstrates that Clothing is the company's strongest-performing category, while young adults represent the most valuable customer segment. The dashboard also reveals opportunities to increase customer subscriptions, improve lower-performing product categories, and enhance overall customer satisfaction.

By combining Python-based data preprocessing with Power BI visualisation, the project successfully transforms raw transactional data into meaningful business intelligence, enabling

managers to make informed, data-driven decisions that support sales growth, customer retention, and operational efficiency.